

DETECT & DEFEND

A 4-DAY INCIDENT RESPONSE & DIGITAL
FORENSICS BLUEPRINT

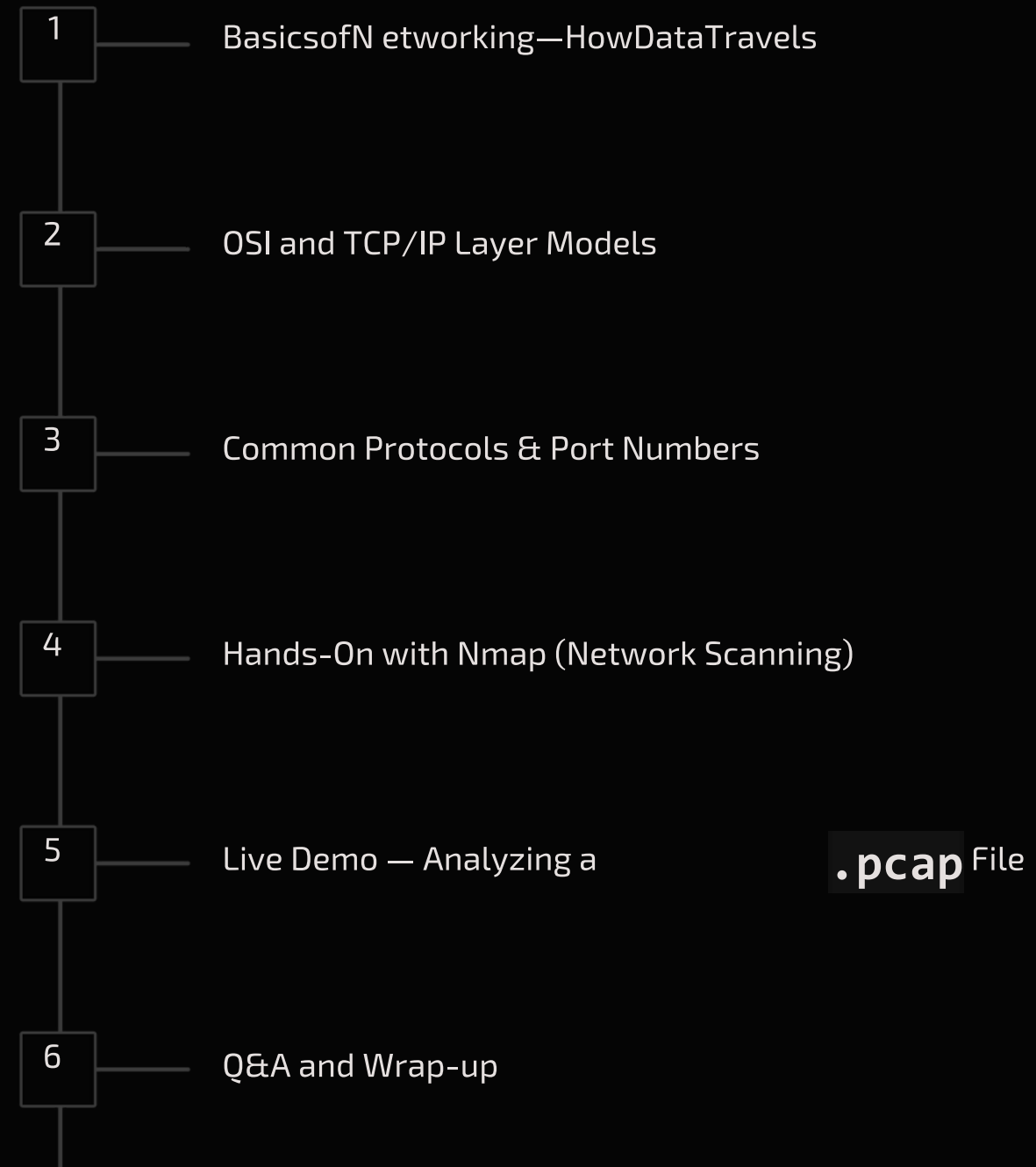


Computer Networking: Understanding How Data
Flows and Tools You Can Use

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Session Roadmap / Agenda



What Happens When You Send Data Over a Network?



The Journey Begins

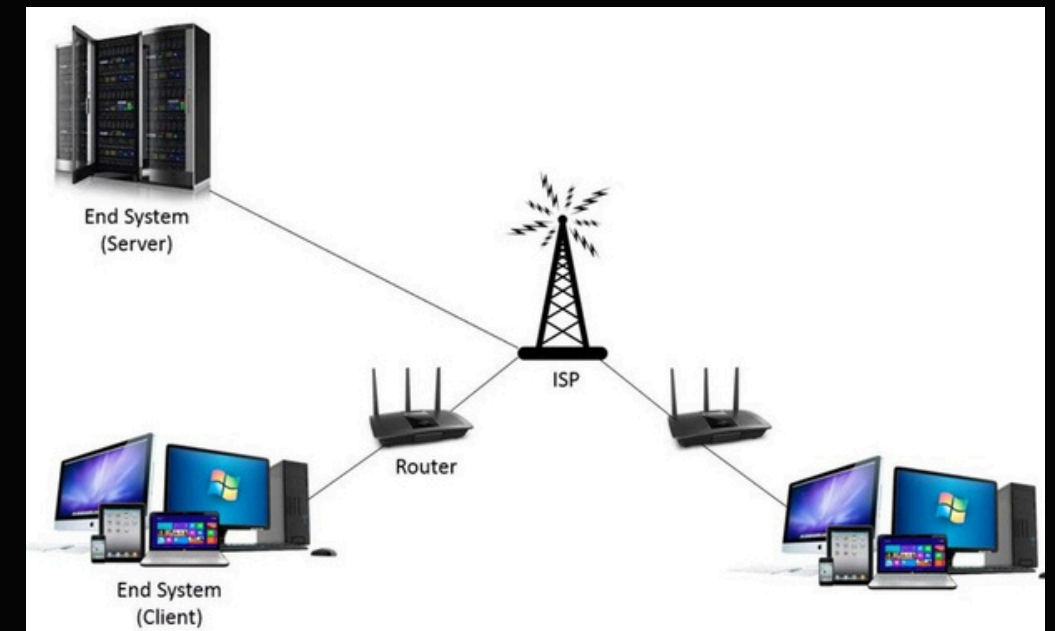
When you click send on a message or load a webpage, your data doesn't travel as one piece. Instead, it's broken into packets - small, manageable chunks that can navigate independently through the network maze.

Each packet contains not just your data, but also addressing information, error-checking codes, and routing instructions.

Hopping to the Destination

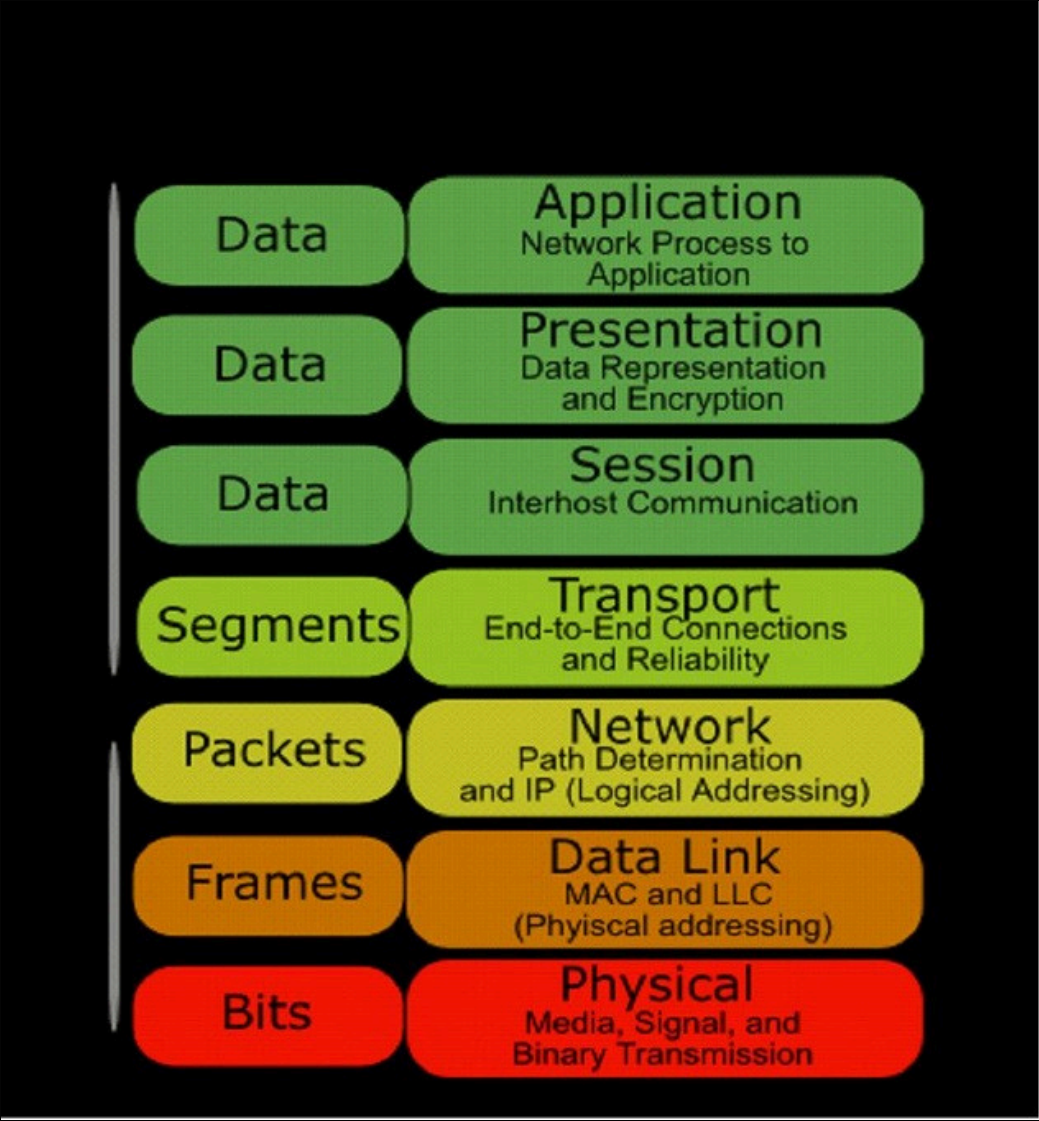
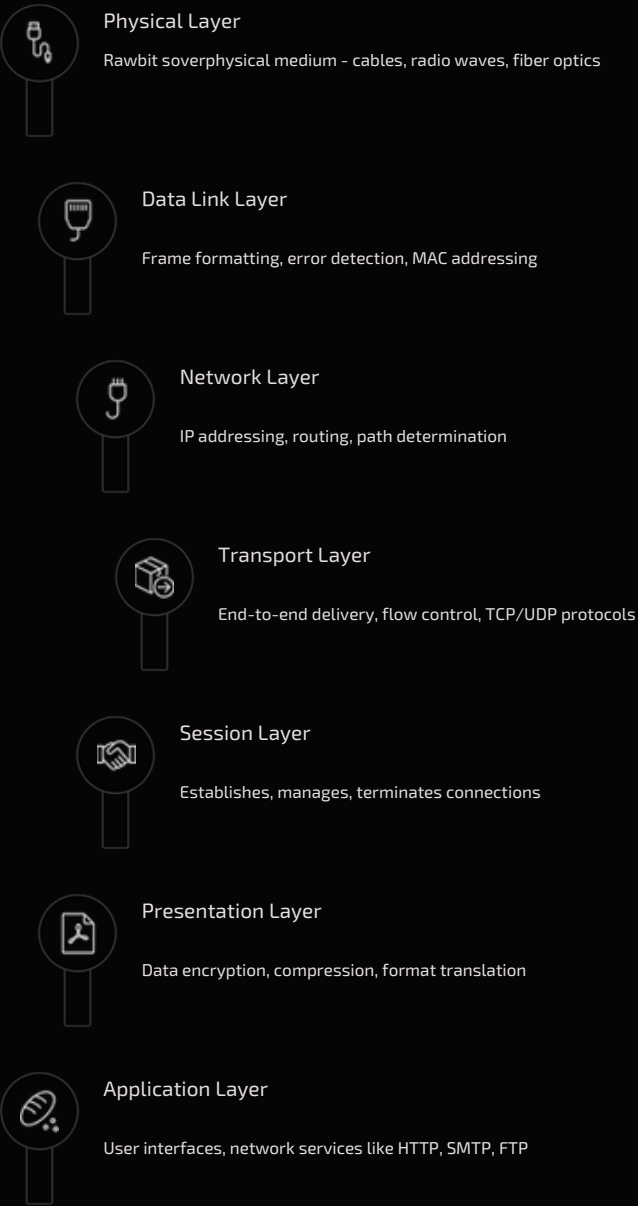
These packets hop from device to device - your router, internet service provider, backbone networks, and finally to the destination server. Each hop examines the packet's addressing information and forwards it closer to its target.

At the destination, packets are reassembled in the correct order, checked for errors, and delivered to the appropriate application. This entire process happens in milliseconds!



The OSI Model: The 7-Layer Blueprint of Network Communication

The Open Systems Interconnection (OSI) model provides a theoretical framework for understanding network communication. Each layer has specific responsibilities, creating a structured approach to data transmission.

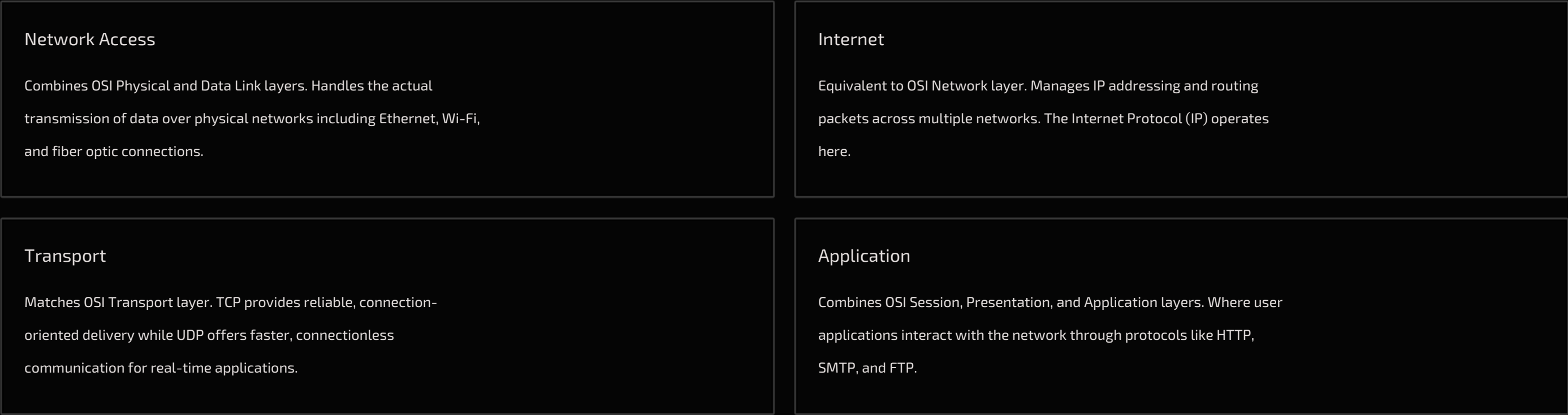


Remember: "Please Do Not Throw Sausage Pizza Away" - Physical, Data Link, Network, Transport, Session, Presentation, Application

The TCP/IP Model: The Internet's Practical 4-Layer Stack



While the OSI model provides theoretical structure, the TCP/IP model reflects how the internet actually works. It combines some OSI layers for practical efficiency and forms the backbone of modern internet communication.



TCP: The Reliable Choice

Transmission Control Protocol ensures every packet arrives and is assembled correctly. Perfect for web browsing, email, and file transfers where accuracy matters more than speed.

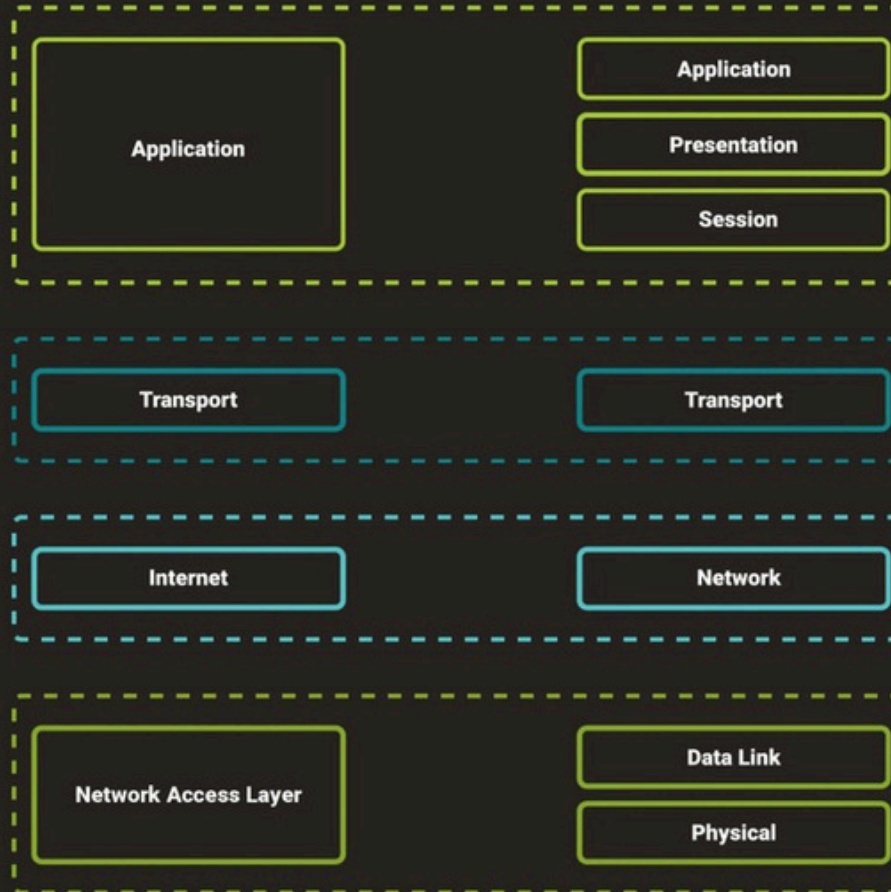
UDP: The Fast Lane

User Datagram Protocol prioritizes speed over reliability. Ideal for live video streaming, online gaming, and DNS queries where occasional packet loss is acceptable.

TCP/IP MODEL

VS

ISO/OSI MODEL

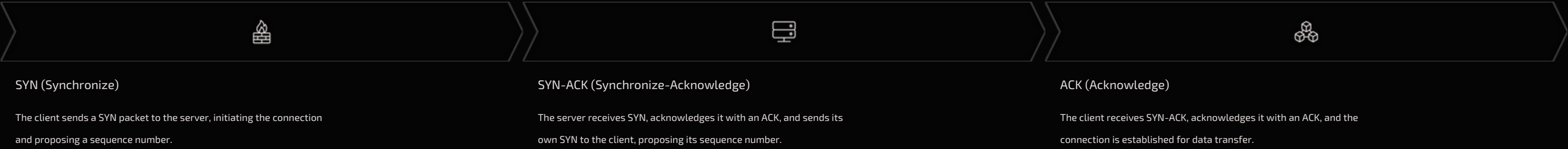


TCP Handshakes: Establishing and Terminating Connections

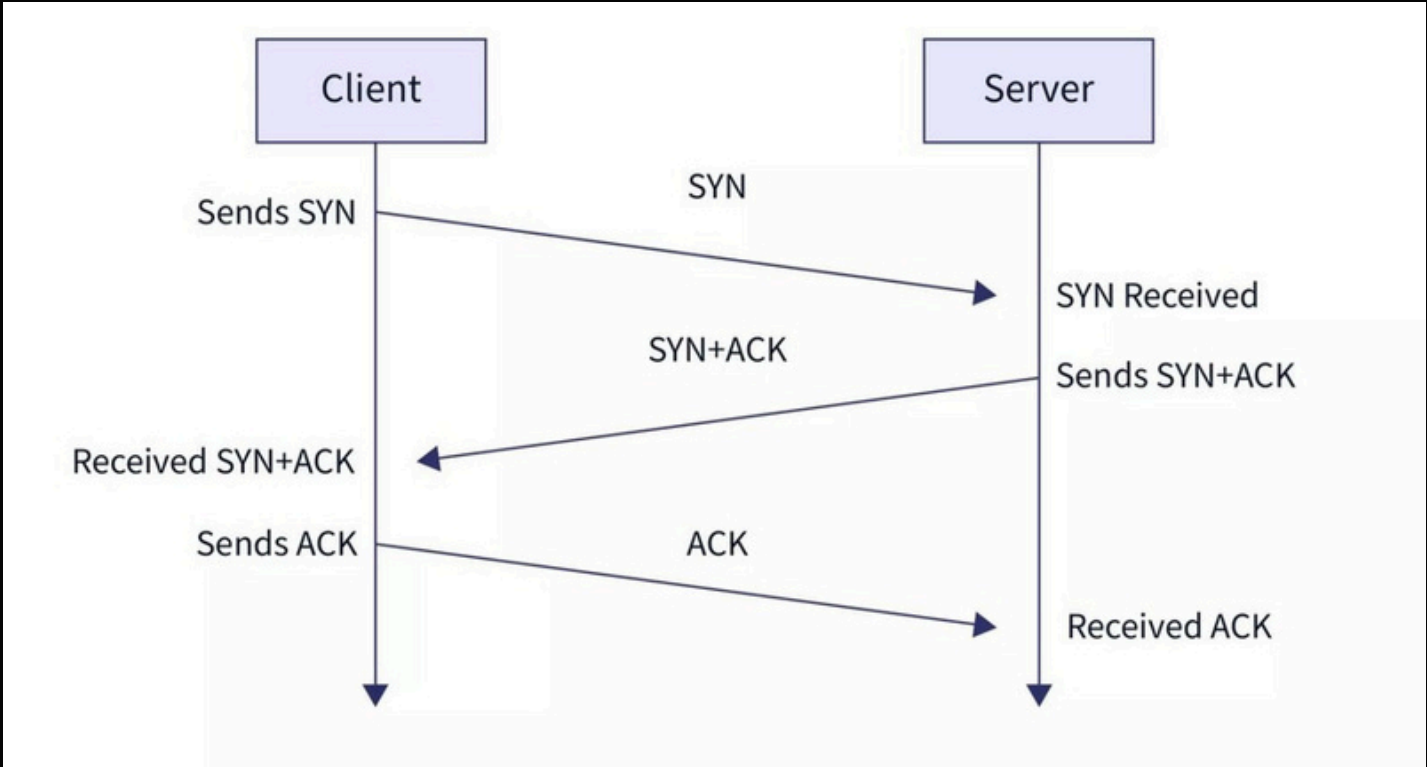


TCP(Tr ansmisionControlProtocol)usesahandshakemechanismtoestablishandterminatereliableconnectionsbetweentwodevices,ensuringbothparties are ready to communicate and gracefully disconnect.

The 3-Way Handshake: Establishing a Connection



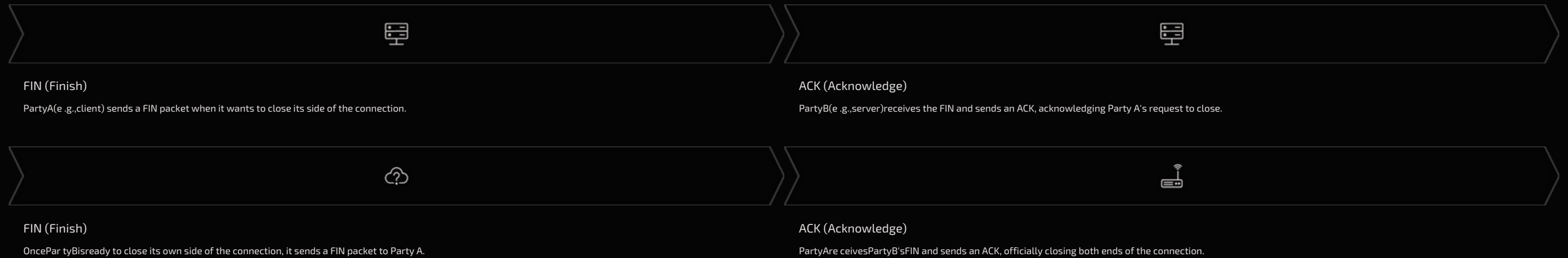
This sequential exchange ensures both sides agree on the connection parameters and are ready to send and receive data.



< 4-Way Handshake >



The 4-Way Handshake: Terminating a Connection



The 4-way handshake allows each side to independently close its transmission channel, preventing data loss during termination.

Key TCP Flags: Beyond the Handshake

Beyond the familiar SYN,ACK, andFIN us ed for establishingandterminatingconnections, TCP segments carry other crucial flags that dictate how data is handled and control the flow of communication. Understanding these flags provides deeper insight into TCP's reliability and control mechanisms.



URG (Urgent)

Signals thatcertain data within the segment is urgent and should be processed immediately by the receiving application, bypassing the normal queuing order.



PSH (Push)

Instructs the s ending TCP stack to immediately "push" all buffered data to the application layer on the receiving end, without waiting for the buffer to fill.



RST (Reset)

Used to abruptly terminate a connection, usually when an error occurs or an invalid segment is received. It indicates that the connection should be immediately closed.

While SYN, ACK, and FIN (discussed previously) manage connection setup and teardown, URG, PSH, and RST handle data delivery priorities and error conditions, ensuring robust communication.

Key Protocols and Their Port Numbers You Should Know

Protocols define the rules of network communication, while port numbers identify specific services running on devices. Understanding common protocols and their associated ports is essential for network troubleshooting and security.



HTTP (Port 80)

HyperText Transfer Protocol powers the World Wide Web. When you visit any website starting with "http://", your browser communicates using this protocol on port 80.



HTTPS (Port 443)

Secure version of HTTP with encryption. The padlock icon in your browser indicates HTTPS is protecting your data as it travels between your device and the website.



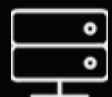
FTP (Ports 20/21)

File Transfer Protocol uses two ports: 21 for control commands and 20 for actual data transfer. Still widely used for uploading websites and transferring large files.



SMTP (Port 25)

Simple Mail Transfer Protocol handles outgoing email. When you hit "send" on an email, SMTP carries your message to the recipient's email server.



DNS (Port 53)

Domain Name System translates website names like "google.com" into IP addresses that computers can understand. It's like the phonebook of the internet.



SSH (Port 22)

Secure Shell provides encrypted remote access to servers and network devices. System administrators use SSH to securely manage systems from anywhere in the world.

How Protocols and Ports Work Together

The Protocol-Port Partnership

Think of protocols as languages and ports as doors. When your browser wants to fetch a webpage, it speaks HTTP (the language) and knocks on port 80 (the door) of the web server.

A single server can run multiple services simultaneously - a web server on port 80, an email server on port 25, and an SSH server on port 22. Each service listens on its designated port, ready to respond to incoming requests.

Port Number Ranges

Well-known ports (0-1023): Reserved for system services and common protocols like HTTP, HTTPS, FTP.

Registered ports (1024-49151): Assigned to specific applications and services by IANA.

Dynamic ports (49152-65535): Available for temporary use by client applications.



Real-world example: When you type "https://github.com" in your browser, you're telling it to use HTTPS protocol to connect to GitHub's server on port 443. The server recognizes this request and responds with the website data encrypted for your security.



Interactive Demo Part 1: Discovering Devices with Nmap

Network Mapper(Nmap)is apowerful,freetoolthatrevealssthehidden landscapeof yournetwork.Itcandiscoverlivehosts, identify open ports, detect operating systems, and even determine what services are running on remote machines.

01

Install and Launch Nmap

Download Nmap from nmap.org and install it on your system. Open a terminal or command prompt to begin exploring.

03

Scan for Open Ports

Target a specific host with `nmap -sS 192.168.1.1` to discover open ports. The -sS flag performs a stealthy SYN scan.

Practical Applications

- Network inventory and asset management
- Security auditing and vulnerability assessment
- Troubleshooting connectivity issues
- Monitoring for un authorized devices

02

Discover Live Hosts

Run `nmap -sn 192.168.1.0/24` to perform a ping scan of your local network. This reveals which IP addresses have active devices.

04

Identify Services

Add `-sV` to detect service versions: `nmap -sS -sV 192.168.1.1`. This reveals what applications are listening on each port.

Ethical Considerations

Important: Only scan networks you own or have explicit permission to test. Unauthorized scanning can be considered a hostile act and may violate laws or policies.

Interactive Demo Part 2: Capturing and Analyzing Traffic with Wireshark



Wireshark transforms your computer into a powerful network detective, capable of capturing and dissecting every packet that flows through your network interface. It's like having X-ray vision into network communications.



Start Capture

Select your network interface and begin capturing live traffic. Watch as packets stream by in real-time, each one telling part of the network story.

Generate Traffic

Open a web browser and visit a website. Wireshark immediately captures the HTTP requests, DNS lookups, and TCP handshakes involved in loading the page.



Apply Filters

Use filters like `http or tcp.port == 80` to focus on specific traffic types. Filters help you find needles in the network haystack.

Analyze Protocols

Click on packets to see their layer-by-layer breakdown. Observe how HTTP headers, TCP flags, and IP addresses work together to deliver data.

What You'll See in Action

- **DNS queries** resolving domain names to IP addresses
- **TCP three-way handshake** establishing connections
- **HTTP headers** containing request details and server responses
- **Packet timing** revealing network latency and performance

Wireshark displays packets in different colors based on protocol type - green for HTTP, blue for DNS, black for TCP errors. This visual coding makes pattern recognition much easier!



Real-Life Applications of Networking Knowledge

Understanding networking fundamentals and mastering tools like Nmap and Wireshark opens doors to solving real-world challenges. These skills are invaluable whether you're a system administrator, security professional, or curious technology enthusiast.

Troubleshooting Connectivity

When networks fail, understanding the OSI model helps you systematically diagnose problems. Is it a physical cable issue (Layer 1), routing problem (Layer 3), or application misconfiguration (Layer 7)? This layered approach saves hours of guesswork.

Enhancing Security

Security professionals use port scanning to identify vulnerable services and packet analysis to detect malicious traffic. Understanding normal network behavior makes it easier to spot anomalies that indicate potential breaches or attacks.

Optimizing Performance

Network analysis reveals bottlenecks, inefficient protocols, and traffic patterns. By understanding how applications use network resources, you can make informed decisions about bandwidth allocation, Quality of Service settings, and infrastructure upgrades.

Career Applications

- Network Engineers: Design and maintain network infrastructure
- Security Analysts: Monitor and protect against cyber threats
- DevOps Engineers: Ensure application connectivity and performance
- IT Support: Resolve user connectivity and access issues

Business Impact

Organizations rely on network professionals to maintain 99.9% uptime, protect sensitive data, and ensure optimal performance for business-critical applications. These skills directly contribute to company success and customer satisfaction.

Summary & Your Next Steps in Networking



Key Takeaways

- Networking follows layered models (OSI and TCP/IP) that standardize communication.
- Protocols and ports work together to enable specific network services.
- Practical tools like Nmap and Wireshark transform theory into actionable insights.
- Understanding networking fundamentals enables effective troubleshooting, security, and optimization.

Your Learning Journey Continues

Set up a home lab with virtual machines to practice scanning and packet capture safely. Experiment with different protocols, create your own network scenarios, and challenge yourself with real-world problems.

Consider pursuing certifications like CompTIA Network+, Cisco CCNA, or specialized security credentials to formalize your knowledge and advance your career.



Build Your Lab

Create a safe environment with VirtualBox or VMware to practice network scanning and analysis without affecting production systems.



Deepen Knowledge

Explore advanced topics like network security, wireless technologies, and software-defined networking to stay current with evolving technologies.



Join Communities

Connect with networking professionals through forums, local meetups, and online communities to share knowledge and learn from experienced practitioners.

Questions & Discussion

Resources

Follow this link to join my WhatsApp community (CyberX Nashik):
<https://chat.whatsapp.com/J03LjPpnU73JflUzzkOiY9?mode=wwt>

And this community for red teaming resources:
<https://chat.whatsapp.com/GcozqJBqnWk4bPQw0zOKBd?mode=wwt>